

Document Revision History

Version	Date	Edited By	Changes Made
V1.0	Initial	—	Original requirements document.
V1.1	2026-06-16	Satyam Kumar	Updated tech stack to match actual implementation: - Backend changed from NestJS / Express to Next.js (App Router API Routes). - Database changed from MongoDB to Supabase (PostgreSQL 17 with Row-Level Security).

Healthcare Appointment Management System

Modular, scalable, and bilingual Healthcare Appointment Management System is organized into three verticals (Patient Mobile App, Patient Web App, and Healthcare Facility App).

Healthcare Facility covers:

- Clinic
- Polyclinic
- Pathology Lab
- Radiology Center
- Hospital
- Physio Clinic
- Skincare & Laser Center

System Structure:

1. Patient App (Mobile)

- Built in Flutter for Android & iOS.
- Designed for easy patient onboarding and booking on-the-go.

2. Patient App (Web)

- Built in NextJs for browser-based access.
- Same features as the mobile app but optimized for desktops/laptops.
- Highly optimized for search engines & GPT.

3. Healthcare Facility (for Doctor, Technician, & Admin Dashboards)

- Built in React.js.
- Includes doctor-facing, technician-facing and facility-facing dashboards for managing appointments, earnings, and patients.

Features:

Feature	Patient App (Mobile)	Patient App (Web)	Healthcare Facility App
Core Patient Management			
Patient profile creation	✓	✓	✓
Patient info management	✓	✓	✓
Family account management	✓	✓	✗

Feature	Patient App (Mobile)	Patient App (Web)	Healthcare Facility App
Patient notes & history access	✓	✓	✓
Doctor, Technician & Facility Management			
Doctor profile (availability, fees)	✓ (view)	✓ (view)	✓ (manage)
Facility profile	✓ (view)	✓ (view)	✓ (manage)
Facility ratings & reviews	✓ (add/view)	✓ (add/view)	✓ (view)
Multi-doctor team calendar	✗	✗	✓
Appointments & Scheduling			
Appointment booking	✓	✓	✓
Appointment rescheduling/cancellation	✓	✓	✓
One-tap rebooking (follow-ups)	✓	✗	✗
Queue management	✗	✗	✓
Web check-in	✓	✓	✓ (View Status)
Calendar sync (Google, iCal, Outlook)	✓	✓	✓
AI-based scheduling assistant	✓	✓	✓
Overbooking control (race condition, buffer time, emergencies, walk-ins)	✓	✓	✓
Multi-location booking for facilities with multiple branches	✓	✓	✓
Payments & Billing			
Digital payment integration	✓	✓	✗
Earnings dashboard	✗	✗	✓
Analytics dashboard (most booked times, patient demographics)	✗	✗	✓
AI Features			
AI doctor suggestion	✓	✓	✗
In-app symptom checker	✓	✓	✗
Communication & Notifications			
Automated SMS/email/WhatsApp reminders	✓	✓	✓
Notifications & push alerts	✓	✓	✓
System & Language Support			
Bi-lingual UI (English & Arabic RTL)	✓	✓	✓

Additional Cost (Estimated):

- Cloud Hosting (Cloudacropolis / ODP): ~ OMR 60–120 / month
- SMS & WhatsApp Gateway Fees: ~ OMR 0.010–0.030 per SMS / message
- Payment Gateway Charges (Thawani / Ompay): Typically 2–3% per transaction
- Email Services (SendGrid / AWS SES): ~ OMR 10–20 / month depending on volume
- App Store (iOS/Android) Developer Accounts: One-time ~ USD 100 (Google) + USD 99/year (Apple)

Deliverables:

- 1 Mobile Patient App – Android & iOS.
- 2 Web Patient App (Next.js) – browser-based patient access.
- 3 Healthcare Facility App (React.js) – Doctor & Admin dashboards.
- 4 Backend (Next.js App Router API Routes) with REST APIs.
- 5 Database (Supabase / PostgreSQL) with proper indexing & backups.
- 6 Integrations: Calendar, SMS/Email/WhatsApp, Payment Gateway.
- 7 Deployment Setup on Cloudacropolis / ODP.
- 8 Documentation: API docs, admin usage guide.

Next Steps:

- **Kick-off Meeting:** Align scope, milestones, and communication protocol.
- **Detailed user journey documents** for each role (patient, doctor, technician, and facility).
- **Agile sprint plan** with timelines, deliverables, and acceptance milestones.
- **Technical architecture & execution plan** with integrations mapped out.
- **UI/UX designs or wireframes.**
- **Design Phase:** Finalize logo, colors, fonts and design artifacts.
- **Development (Agile Sprints):** 8-week sprint cycle with weekly demos.
- **UAT & Feedback:** Client reviews and approve deliverables.
- **Deployment:** Deploy Cloudacropolis/ODP and publish mobile apps.